

# SmartFarm

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**An offline farm-management app  
built for Cambodian small-scale  
farmers.**

Track money. Plan activities.  
Identify pests. All in Khmer. All offline.

Presented by the SmartFarm team

## THE PROBLEM

# Farming small plots is a daily juggle of memory and guesswork

### Untracked money

Income and expenses live in notebooks or memory.  
By harvest time, no one knows real profit.

### Missed activities

Planting, watering, spraying, harvesting — easy to lose the schedule.  
A missed window costs a season.

### Unknown pests & disease

By the time damage is visible, the right treatment  
is often guessed at or asked from neighbors.



### Patchy connectivity

Rural areas have weak or no internet.  
Cloud-only farming apps don't work where farmers actually farm.

# One offline app, four tools farmers already need

## Finance Tracker

Income & expense in KHR or USD.  
Live balance.  
Categories, filters, dual currency.

## Calendar & Activities

Month grid with activity dots.  
Plant / Tend / Protect / Harvest types.  
Mark done, never forget.

## Pest & Disease Guide

Bundled offline library.  
Search + category filters.  
Symptoms, treatment, prevention.

## Settings & Backup

Light / dark theme.  
CSV export, JSON backup & restore.  
One tap to clear & start fresh.

100% offline · Khmer-first UI · KHR + USD · No accounts, no data leaves the phone

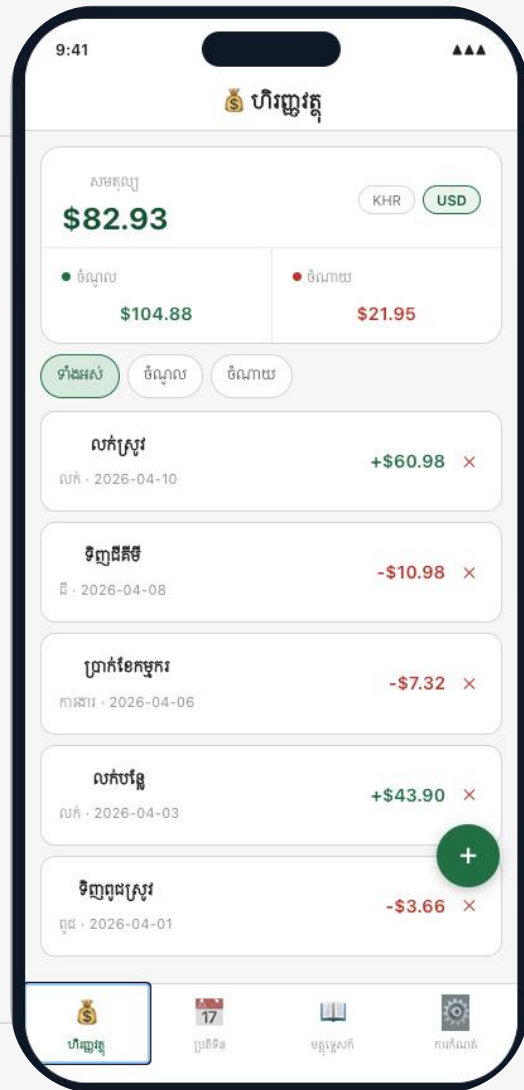
# From first launch to daily use



**First run takes ~30 seconds. Daily use is single-tap actions.**

Persistent local storage means the app opens with your data — every time, online or off.

# Finance Tracker — track every riel and dollar



## Why farmers care

Profit isn't profit until you've counted it.  
Most small farmers know revenue but not net.

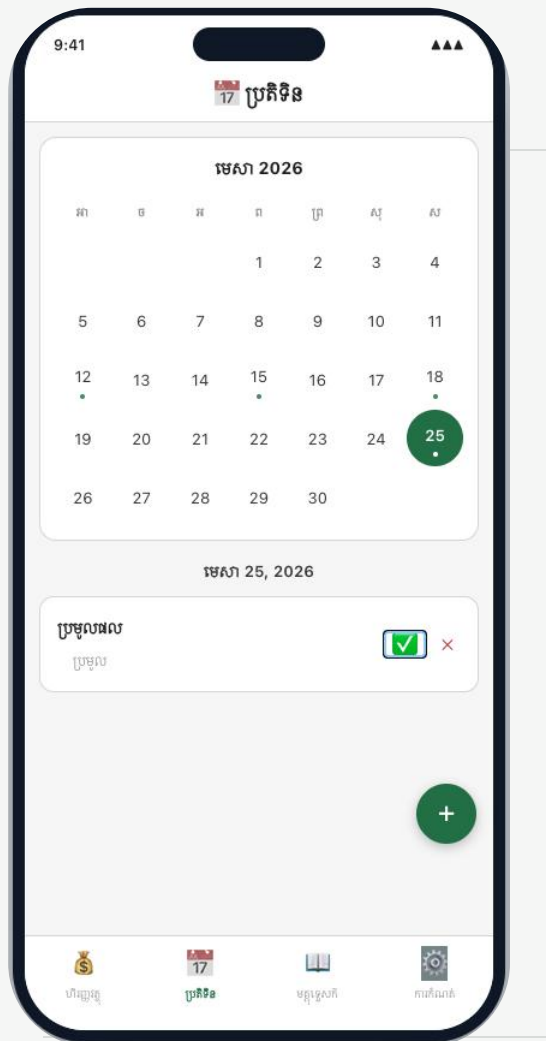
## What it does

- Add income / expense in seconds (FAB → bottom sheet)
- Dual currency: KHR ↔ USD at a stable rate (4,100)
- Categories: ដី, ពូជ, ការងារ, ឧបករណ៍, លក់
- Live balance, income, expense breakdown
- Filter chips: All / Income / Expense
- CSV export for record-keeping or microfinance applications

### Add Transaction Flow

Tap + → Income / Expense → Amount + Date + Currency → Category + Note → Save

# Calendar & Activities — never miss a season



## Why farmers care

Crop cycles are unforgiving.  
A missed spray window or a late harvest costs an entire season's margin.

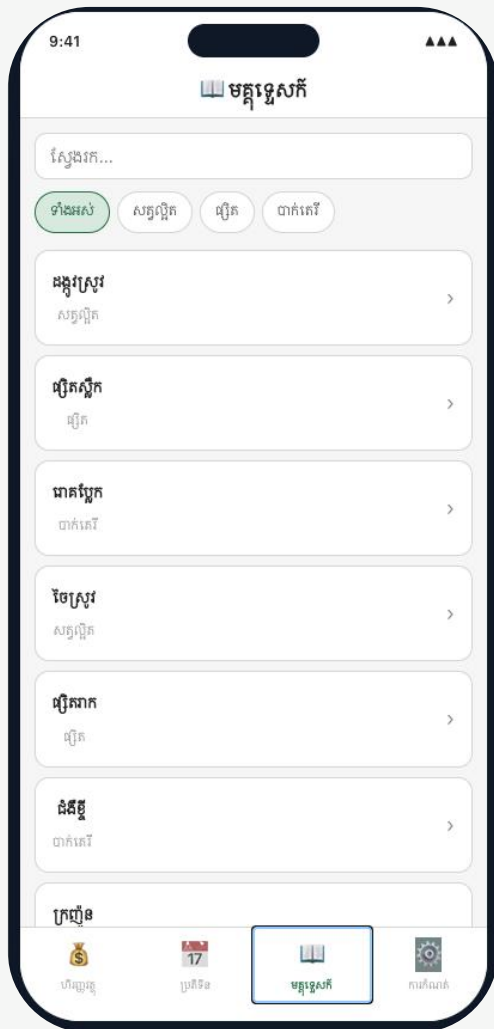
## What it does

- Month-grid view with dot markers on busy days
- Activity types: ដំណាំ (Plant) • ថែទាំ (Tend) • ការពារ (Protect) • ប្រមូល (Harvest)
- Tap a day → see only that day's activities
- Mark done with a single tap
- Add → Name + Date + Type → save → calendar dot appears

## Schedule Activity Flow

Pick day → Tap + → Name + Date + Type → Save → Dot on calendar

# Pest & Disease Guide — answers without internet



## Why farmers care

When a pest appears, decisions are minutes — not days.  
Online services aren't an option in the field.

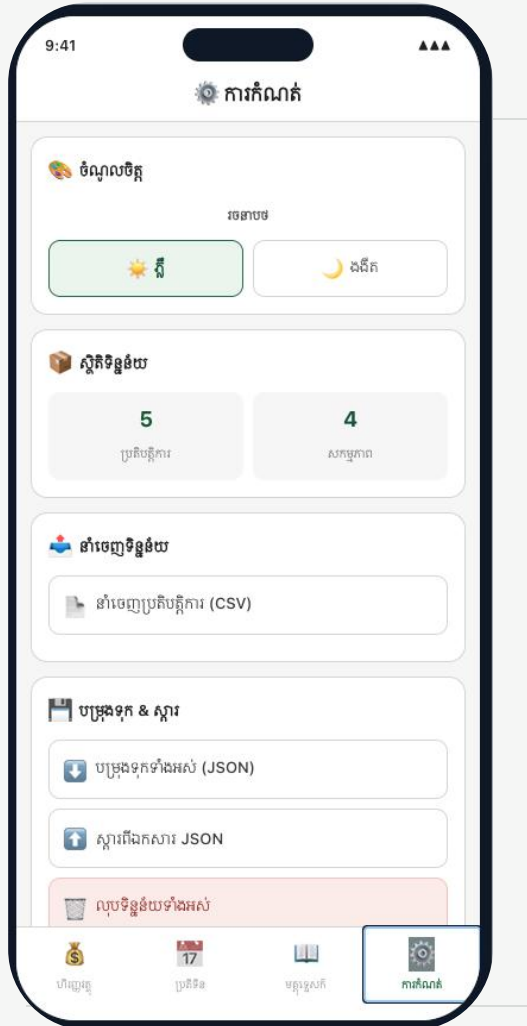
## What it does

- Bundled offline library — no network call ever
- Search by Khmer name
- Category filter: សត្វល្អិត (Insects), ផ្សិត (Fungal), បាក់តេរី (Bacterial)
- Detail view with expandable sections:
  - រោគសញ្ញា (Symptoms) • វិធីព្យាបាល (Treatment) • វិធីការពារ (Prevention)
- Designed to scale to 30–50 entries with photos

## Lookup Flow

Search or Filter → Tap pest → Expand sections → Apply treatment

# Settings & Backup — own your data



## Why farmers care

Phones break. Phones get sold. Phones get reset.  
Two seasons of records vanish in one moment without backup.

## What it does

- Light / dark theme — usable in bright sun and at night
- Data stats — see counts at a glance
- CSV export — share with cooperatives, MFIs, tax officers
- Full JSON backup — one tap, save to Files / share / cloud
- Restore — load a backup file, replace current data
- Clear all — fresh start, with a confirmation guard

### Backup & Restore Flow

Settings → Backup all (JSON) → Save file → (later) Restore from file → Confirm

# Four reasons SmartFarm is different

**[OFF]**

## Truly offline

Every feature works with airplane mode on.  
Reference data is bundled. Storage is local.  
Cloud-first competitors fail in rural Cambodia.

**[KH]**

## Khmer-first by design

Every label, every category, every pest name —  
in Khmer. Not a translation layer.

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## Dual currency native

KHR and USD both first-class.  
Reflects how Cambodian commerce actually works.

**[--]**

## No account, no friction

Open the app. Use it.  
No login, no email, no data leaves the phone.

# Working demo today, native iOS tomorrow

## Today

- Working React + Vite demo
- Khmer UI, KHR/USD, theme toggle
- All four modules functional
- CSV export + JSON backup live

## Next 3 months

- Native iOS port (Swift / SwiftUI)
- iOS 13+ deployment target
- Photo-based pest detail (30–50 entries)
- CoreData persistence
- Local notifications for activities

Stack: React 18 · Vite · localStorage → SwiftUI · CoreData · iOS 13+

# A real audience, a real product, a clear path

## Cambodia

### agriculture-led economy

Small farmers form the backbone of rural livelihoods.  
Few have any digital tool today.

## Smartphones

### wide & growing reach

Entry-level older iPhones are common, even in rural districts.

## Connectivity

### still the blocker

Network coverage is uneven; data is costly.  
Offline-first is the unlock.

### What we have right now

- A working app, not a deck — every feature on the previous slides is built and runs offline today.
- Khmer UX validated against the real domain (categories, activity types, common pests).
- A clear, conservative roadmap to a native iOS app supporting devices farmers actually own.

# Why farmers will pick SmartFarm over what they have today

| Feature                         | Paper notebook<br>(today) | Cloud farming app<br>e.g. Plantix | SmartFarm<br>(us) |
|---------------------------------|---------------------------|-----------------------------------|-------------------|
| Works fully offline             | ✓                         | ✗                                 | ✓                 |
| Khmer-first interface           | ✓                         | ✗                                 | ✓                 |
| Searchable records              | ✗                         | ✓                                 | ✓                 |
| Built-in pest & disease library | ✗                         | ✓                                 | ✓                 |
| Activity reminders & calendar   | ✗                         | ✓                                 | ✓                 |
| Backup & restore                | ✗                         | ✓                                 | ✓                 |
| Free, no account required       | ✓                         | ✗                                 | ✓                 |
| Designed for Cambodian farmers  | ~                         | ✗                                 | ✓                 |

SmartFarm is the only option that wins on every axis rural Cambodian farmers actually care about.

# Thank you

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Questions, partnerships, pilots — let's talk.

SmartFarm — កសិកម្មឆ្លាតវៃ ងាយស្រួល